

Human Scream Detection

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Abstract—Crime, assaults, and robberies is a constant occurrence on a global scale, making it a major worry in our society. Typically, police arrive at crime sites too late, which is a regular problem. This is typically caused by a lack of access to up-to-date and correct information. A disguised desktop application is proposed as a possible approach to alleviate this concern. The program uses modern technologies, such as machine learning and deep learning models like Support Vector Machines (SVM) and Multilayer Perceptron (MLP), to detect and analyze human sounds quickly while operating quietly in the background

Index Terms—Crime rate, Scream detection, SVM, MLP.

I. INTRODUCTION

Crime has been a problem all societies have been deal for years if not centuries throughout history. Given the rising frequency of criminal acts such as physical assaults, thefts, and homicides, preventative measures must be implemented to mitigate their harmful consequences. One key issue that points the root of the problem is the delayed law enforcement response times, which can be caused from many things, but we are going to focus on the lack of reliable information. This project aims to solve what we thin might be one the biggest reasons crime has been rising nowadays. Controlling crimw rates using machine learning and deep learning techniques we want to address the main problem.

II. METHODOLOGY

This study shows a new approach to fight crime using a sophisticated tool as machine learning and deep learning algorithms, more specific, Support Vector Machine and Multilayer Perceptron Neural Network. The main goal is to develop a system able of real time detection and analysis of human screaming.

The SPV was the initial method employed in our model. The primary purpose of SVM is to detect human screams within the audio data. SVM has high proficiency in categorizing and discerning positive and negative class instances following training on a selceted dataset. The positive class contains 2.000 instances of human screams while the negative has about 3.000 instances of non-scream sounds. This binary dataset ensures that the SVM model acquieres all the neccesary information to accurately detect genuine threats.

The MPN is a secondary approach we used to work with the SVM model. The MPN serves as a validation technique to enhance the system's ability to detect human screams. The MPN improves the reliability of the system by providing a more comprehensive representation of audio data on the

identical dataset. MPN trives at recognizing intricate patterns due to its capability in deep learning.

Using the LibRosa tool we extracted the Mel Frequency Cepstral Coefficients MFCCs from the dataset was an big part of our project. Since MFCCs are a vital part if you want to fully understand speech and audio data, libRosa helped us to improve the performance on both SVM and MNP models.

TensorFlow is a heavy machine learning platform used for storing and executing SVM and MPN models. The integration assures a great collaboration between the detection methods and broader systems without facing any major issues. This technique combines two models, demonstrates its effectiveness by incorporating features like MFCCs.

$$f(x) = \text{sign}(w * x + b) \quad (1)$$

The equation represents the decision function for SVM. Where:

x : Input feature
w : Weight vector.
b : Bias term and sign

- **Input:** Voice examination
- **Phase 1:** Thje goal of outdoor noise filterin is to eliminate or minimize auditory disruptions caused by background noise.
- **Phase 2:** classifies the sound into different categories
- **MLP:** The input interpreted as possible scream is transmittted to a multilayer perceptron model for a more thorough process.
- **Decision:** If the MLP model detecs a scream, it makes warning

III. RESULTS AND ANALYSIS

The SVM and MPN model demonstrated an accuracy of 75.5% on recognizing human screams in uploaded audio files giving a notification of "Scream Detected" or "No Scream Detected" as we can see in Figures 1 and 2.

IV. CONCLUSION

Both images represent the output of our project when an audio file is uploaded. AS a future work we are planning on adding a real time classifier and a notification sending a SMS when a scream is detected.

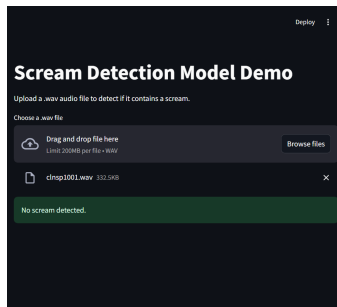


Fig. 1. No scream detected

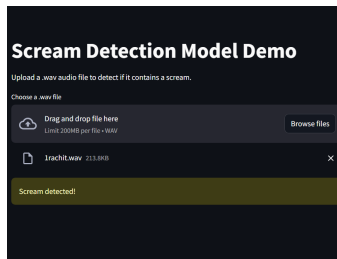


Fig. 2. Scream detected

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